



Medical Microbiology

Third stage- College of Dentistry

Lab. 12

Mycobacterium tuberculosis

Prepared by:

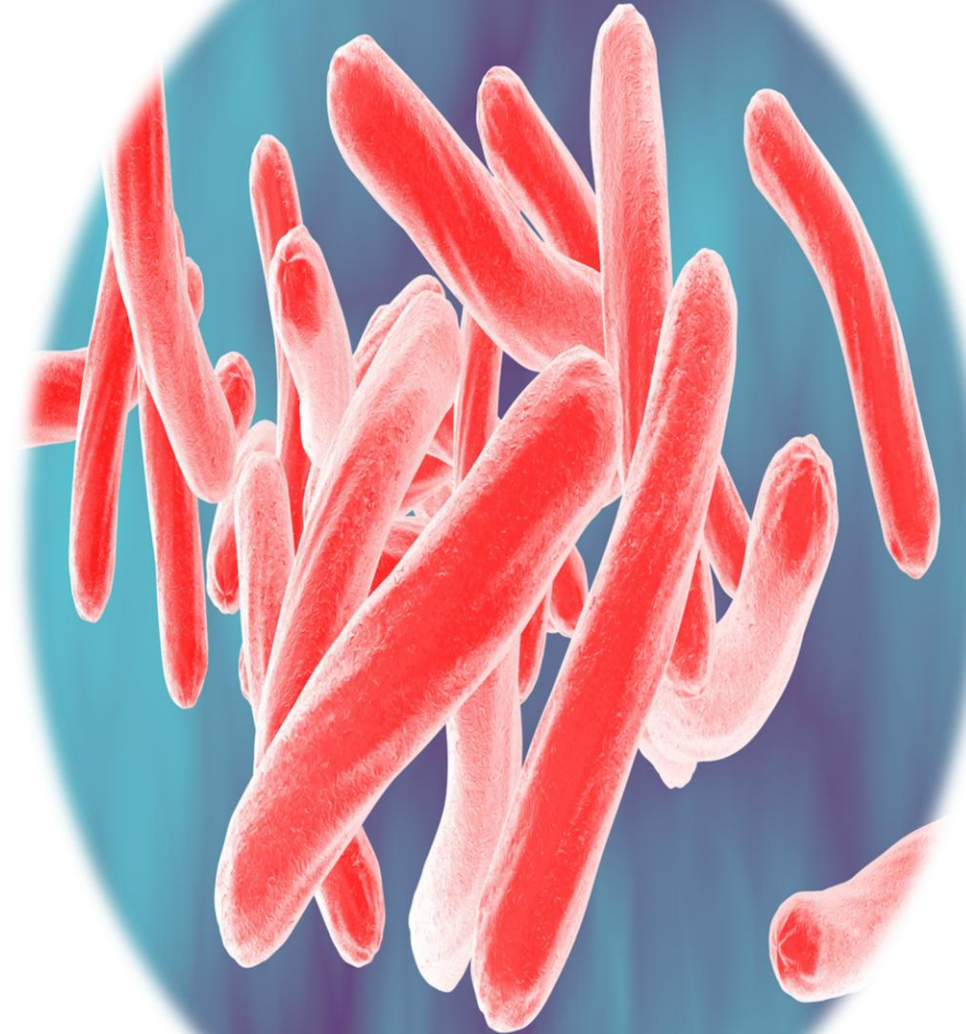
PhD. Dheyaa Naji Hamza

M.Sc. Zainab Majid Mohammed

Learning Objectives:

After completing this lab, you should be able to:

- 1-Describe the microscopic appearance of *Mycobacterium tuberculosis* .
- 2-Explain the laboratory methods used to detect *Mycobacterium tuberculosis* .
- 3-Recognize its cultural characteristics and the selective media used for its isolation.



Characteristics of *Mycobacterium tuberculosis*

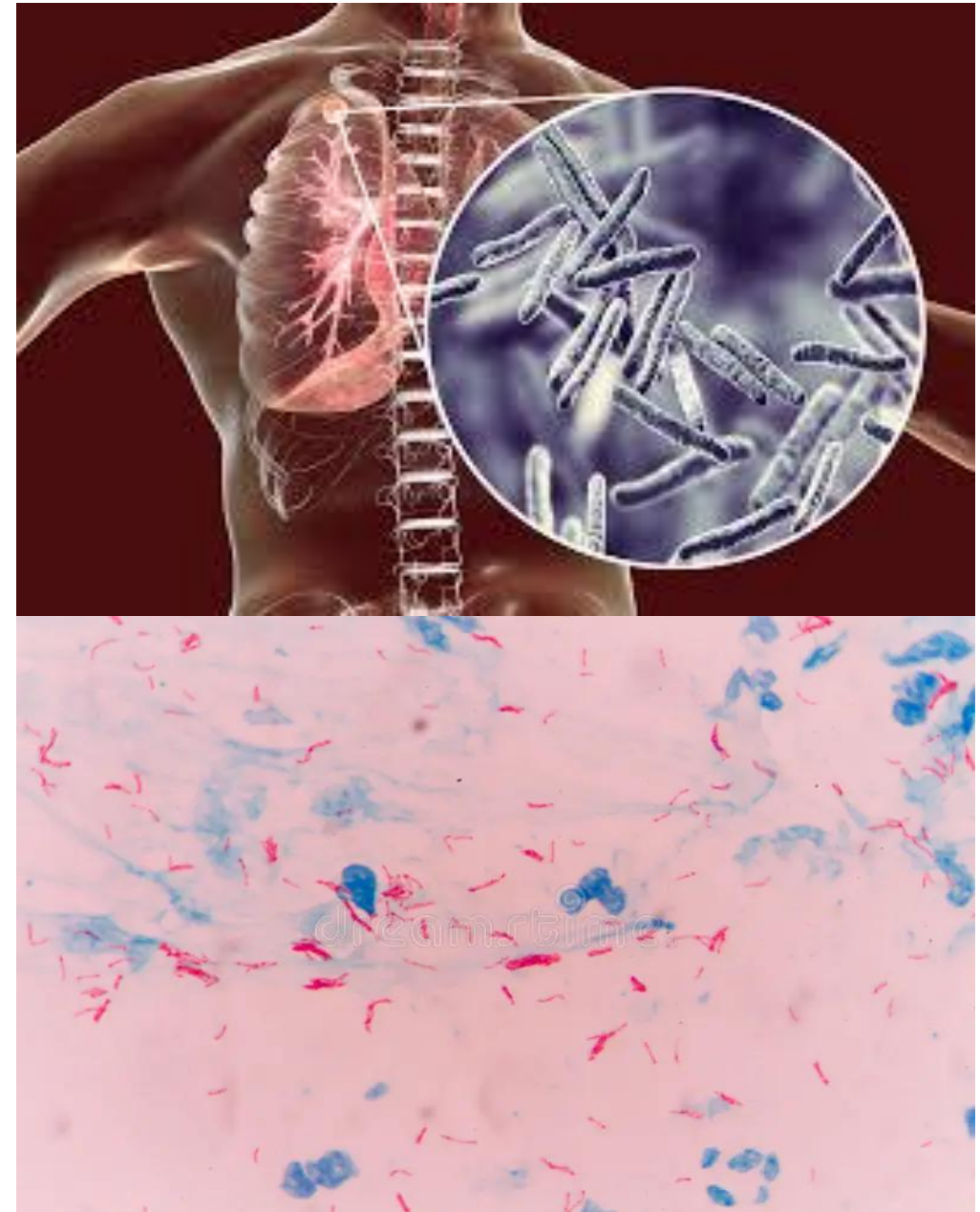
Shape and Size: Rod-shaped, Slender, straight, typically 1 to 4 micrometers in length.

Respiration: Obligate aerobe.

Motile: Non-motile.

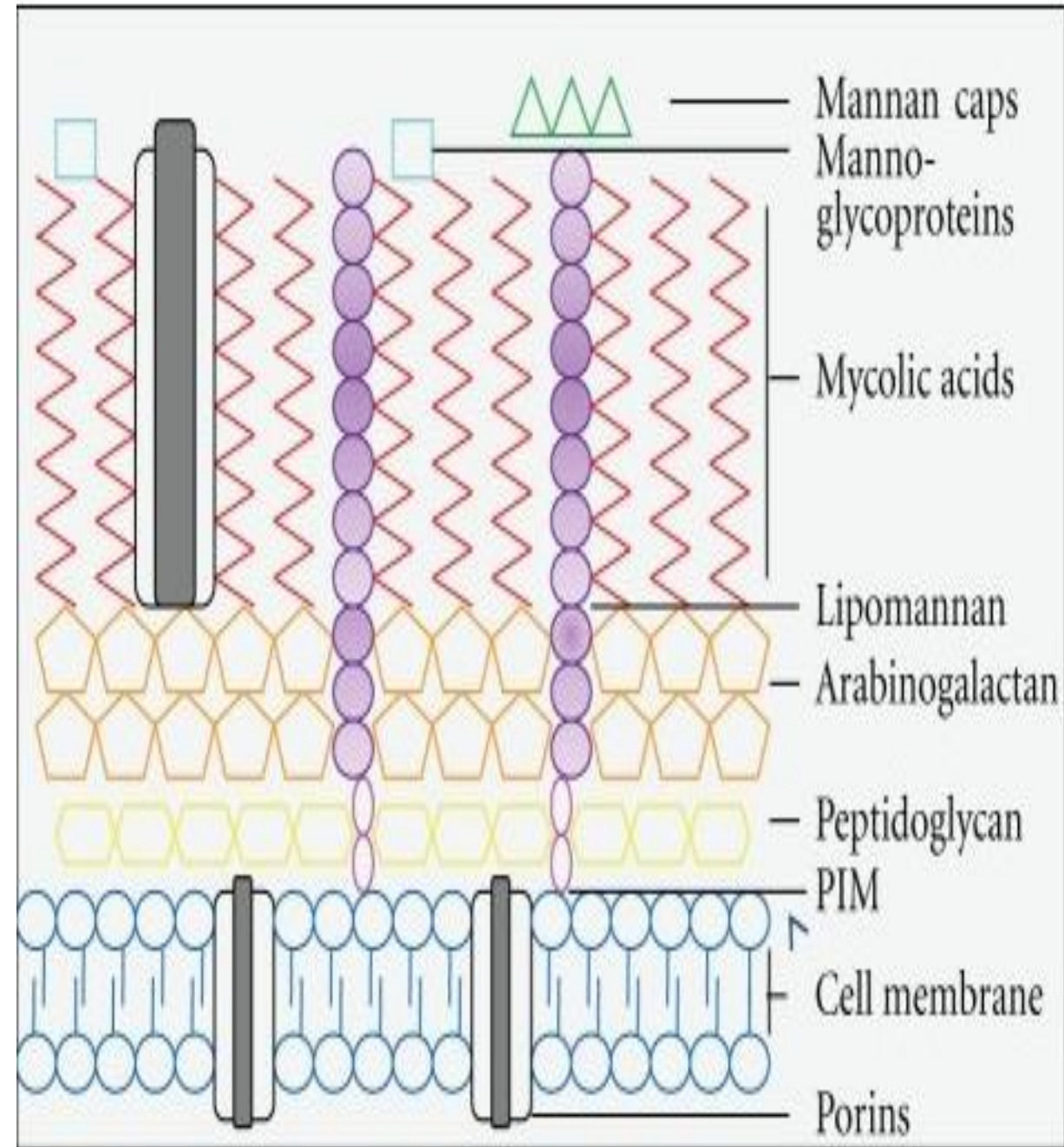
Spore: Non-spore forming.

Capsulated: Non-capsulated.



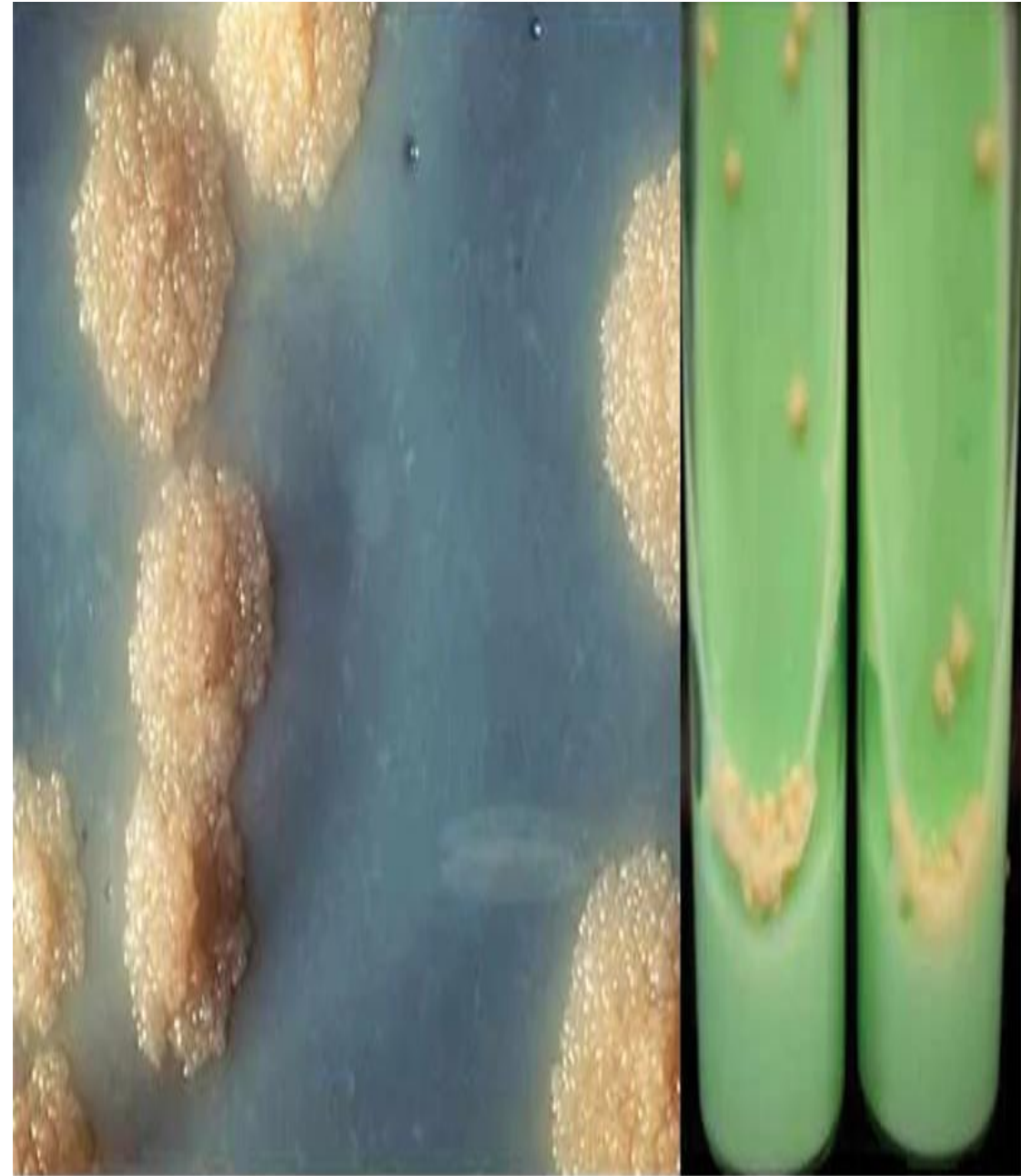
Cell wall structure:

- contains **peptidoglycan** and complex **lipids**.
- Over 60% of the mycobacterial cell wall is lipid.
- The lipid fraction of MTB's cell wall consists of three major components, mycolic acids, cord factor, and wax-D, which makes it resistant to ordinary stains like Gram stain, and also resistant to common disinfectants.



Selective media: Lowenstein-Jensen medium It is an egg-based medium prepared as slants, containing eggs, potato, serum, glycerol, malachite green, and antibiotics.

Colonies appear dry, rough, wrinkled; these bacteria are very slow growers it takes 4-6 weeks until visible growth appears.



Method of Staining TB Bacteria (Ziehl-Neelsen Stain):

TB bacteria require a special staining method known as the Ziehl-Neelsen stain (or "Acid-Fast Stain").

Principle

This method relies on **heat** to melt the waxy components of the bacterial cell wall, enabling the **primary stain (carbol fuchsin)** to penetrate the cells. As the slide cools, the wax re-solidifies, **locking the stain inside**. When **acid-alcohol** is applied, only organisms that retain the stain—such as *Mycobacterium tuberculosis*—**resist decolorization** and remain stained.

Required Materials

Primary Stain: Carbol-fuchsin.

Decolorizer: Acid-alcohol (usually 20% alcohol with hydrochloric acid).

Counter stain: Methylene blue or malachite green.

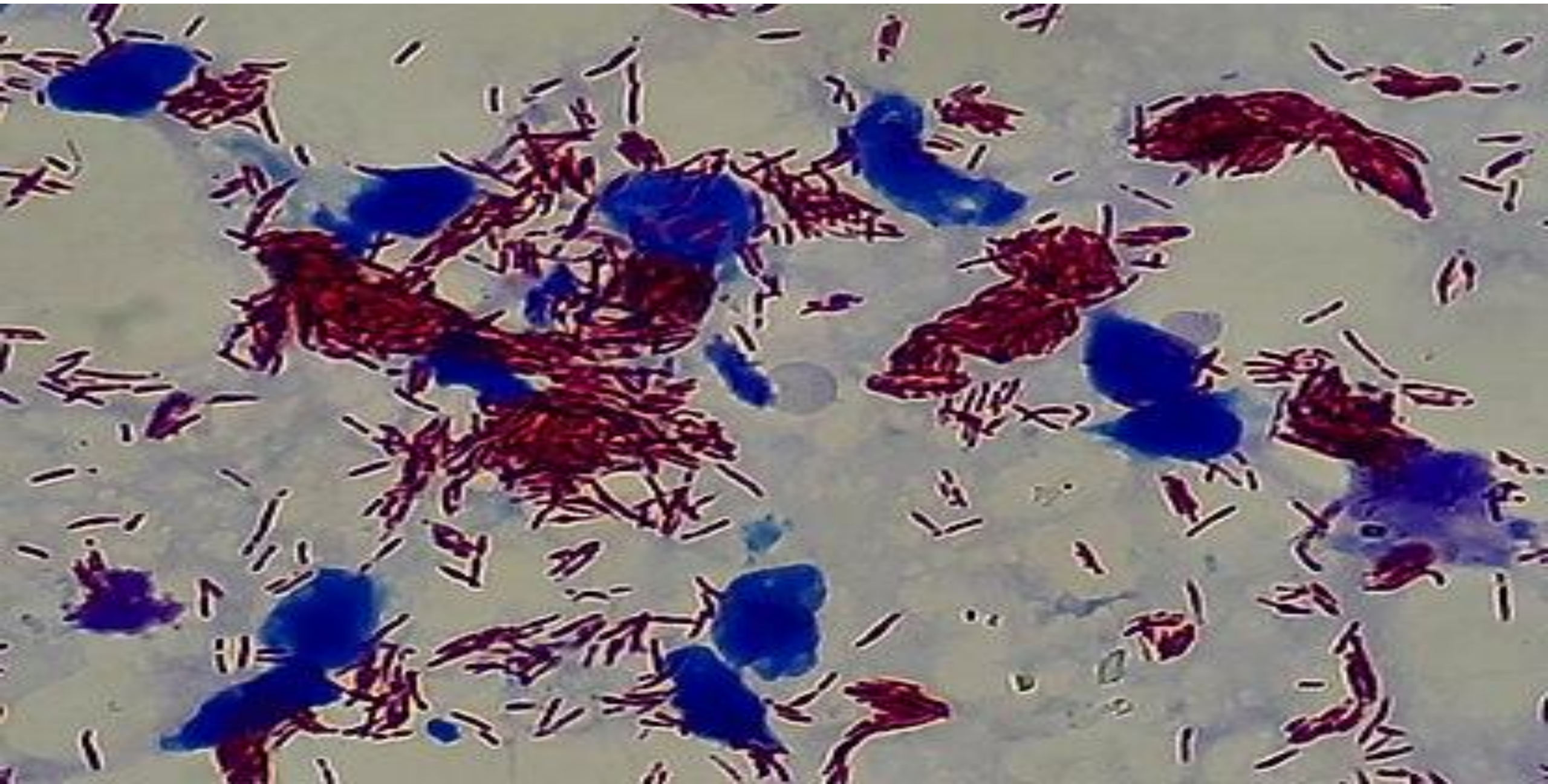
Procedure:

- 1-Prepare a thin smear on a clean glass slide, air-dry, and heat-fix.
- 2-Flood the smear with **carbol fuchsin** and gently heat until steam appears (do not boil); keep for 3–5 minutes.
- 3-Rinse the slide with water.
- 4-Decolorize with **acid-alcohol** until no more color runs off.
- 5-Wash with water.
- 6-Counterstain with **methylene blue** for 1–2 minutes.
- 7-Rinse, air-dry, and examine under oil immersion.

Result:

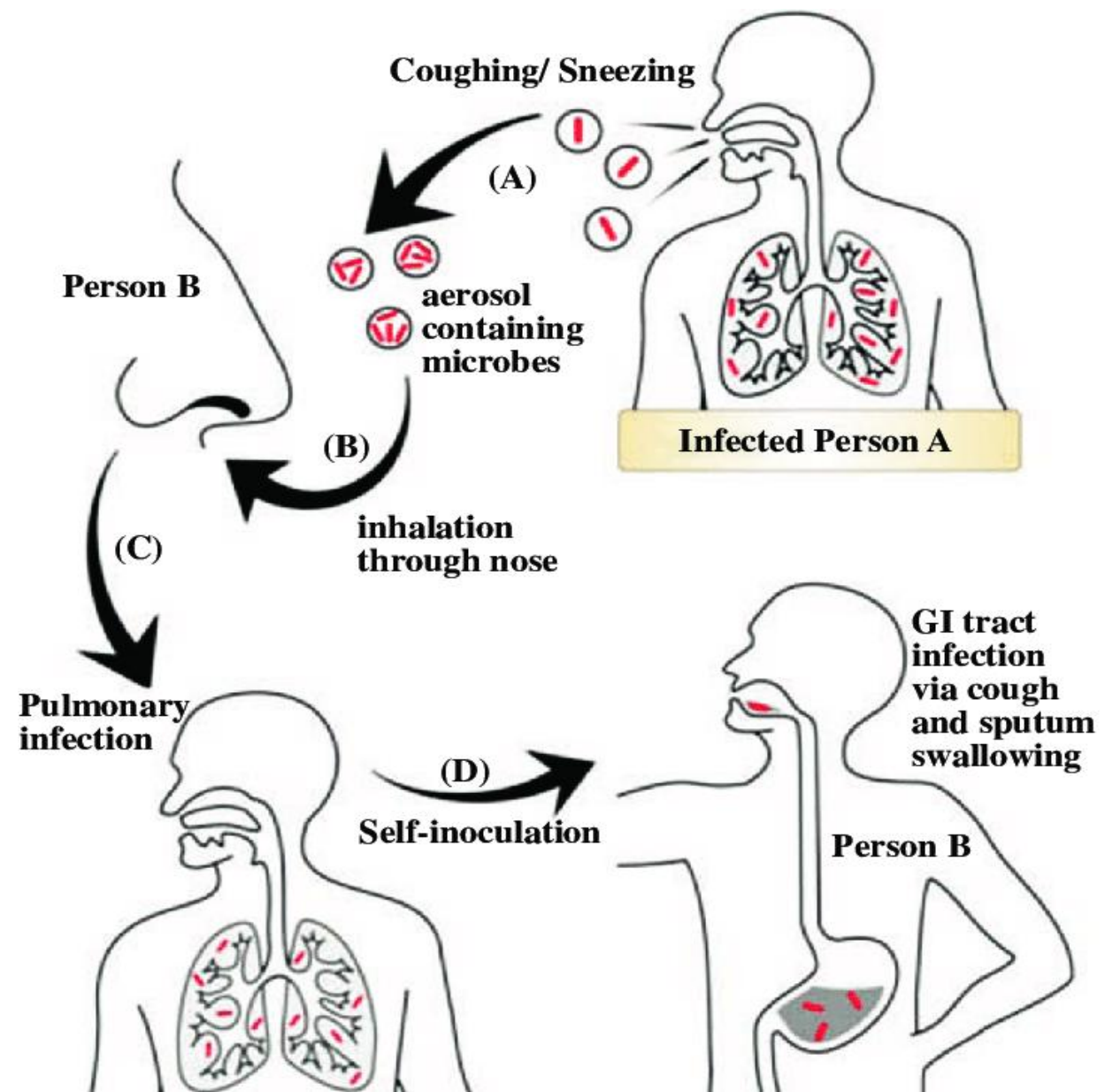
Acid-fast bacilli (*Mycobacterium tuberculosis*) appear **red**, while background cells appear **blue**.

M. Tuberculosis (Ziehl-Neelsen Stain)



Transmission: Primarily transmitted through the air when an individual with active pulmonary TB coughs, sneezes, or talks.

Pathogenicity: Primarily infects the lungs, but can attack any other part of the body (such as bones, kidneys, digestive system). The bacteria can survive inside macrophages in the lung without being destroyed.



Symptoms of active TB disease:



Cough lasting
3+ weeks



Coughing up blood or
sputum (*phlegm from
deep inside the lungs*)



Chest pain



Weakness
or fatigue



No appetite



Weight loss



Fever and/or chills



Night sweats



Any Question?



Thank you